Introduction

Setup

Ask two C47 or R47 users to set up the calculator and you'll get two different setups. Ask three and you might start an argument. The defaults are perfectly reasonable. The problem is that "reasonable" is a personal idea, and after a week with the calculator you start having opinions about it.

Opinions

The mathematician next door wants to see 2π, not 6.283. The astronomer is happy with the default exponent range. The accountant wants to know who on earth needs 10^{6145}. The mechanical engineer wants fractions to land on sixteenths. The financial analyst doesn't want fractions at all, thank you. Some have a view on whether the imaginary unit should be i or j, and nobody is wrong.

Aim

The C47 and R47 firmware lets all these people have what they want. This note isn't really about saving your preferences as a program (Application Note 15 covers that, and a topic for once you're properly settled in). Most people, in our experience, just want to know where the switches are so they can set the calculator up the way they like it. So that's what this note is: a light tour of where the settings live, with particular attention to the regional differences that catch people out, followed by a quick walk through what different disciplines do with all of it.

Three layers

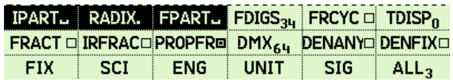
Layers

The configurability story makes more sense if you picture it as three layers rather than one big pile of switches. The layers correspond to how often you actually visit them.

Display

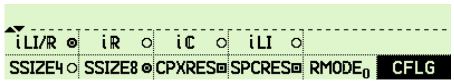
The first layer is reached through DISP, and it's the one most users learn first. This is where the calculator decides what numbers and decimals look like. FIX, SCI, ENG, UNIT, SIG, ALL. How many digits each carries. The radix character. The separators on either side of it. Fractions and irrationals, so that 2π and √2 can appear as themselves rather than to be half-rounded decimals. Fraction maximum denominator, fractions mixed or improper. Complex numbers get i or j, and the multiplier can be shown or hidden. There are also regional presets bundled in here, and they're important enough that they get their own section below. Most users use in this layer most of the time.

Taste



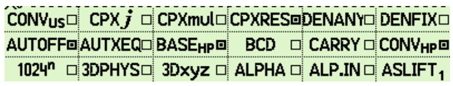
Display mode is the single area that users hold the strongest opinions about, and there is no consensus. FIX is the calculator showing its working with a steady number of decimals, which is what most people want for money, dimensions, and anything that needs to line up in a column. SCI is what physicists and astronomers reach for, because the exponent is doing the heavy lifting and the mantissa is just along for the ride. ENG is the engineer's natural habitat, jumping in steps of three so the prefix follows. SIG counts significant figures, which is what laboratory work, field work, and frankly the world out there actually requires. UNIT attaches an SI prefix to the value. ALL shows everything the calculator has. None of these is the right answer; they are six different answers to six different questions, and one of the small pleasures of the C47 and R47 is that you can change your mind in one keypress.

Behaviour



The second layer concerns behaviour rather than appearance. Stack size. Classic or entry RPN. The rounding mode. Whether complex results are allowed. Whether ∞ and NaN pass through quietly or trip an error. The default input datatype. Solver precision. Also the UI behaviour like f/g lines and HOME/MyMenu behaviour are set here. Often users fiddle with and set this layer once when they get the calculator and then leave it alone.

Flags



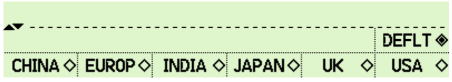
The third layer is the flag layer. This is where the firmware exposes the individual switches that the rest is built on. Plotting markers. Status bar visibility for fractions, integer mode, stopwatch, date, time. Date format. 24h or 12h. Spherical coordinates in the Physics or Math convention. xyz or ijk for vector display. Bit-level flags for carry, overflow, BCD. Long-press behaviour. Printer modes for those who still have a printer. This layer rewards a careful afternoon of poking around when you first get the calculator. After that it doesn't need much.

In short

DISP for what numbers look like, the system preferences for how the calculator behaves, and the flag menu for everything else. The common things appear in more than one place. The full list of flags lives in the third layer, but the ones people change often have buttons in the first two layers too. The firmware is being polite about it.

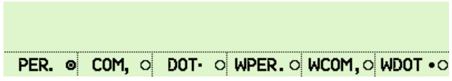
Regional differences

Region



The single biggest split between users is regional, and it deserves a proper look. The calculator can be retuned for a handful of regions in one keypress, but the underlying choices are worth understanding because they catch people out even after years of use.

Radix



The decimal point is not a universal idea. In the US it’s a dot, with commas as thousand separators: 1,234.56. The UK is moving towards the SI recommendation of a dot with thin spaces (or nothing) for grouping: 1 234.56. Most of Continental Europe uses a comma as the decimal, with dots or spaces for grouping: 1.234,56 or 1 234,56. In Switzerland the convention is a dot with an apostrophe as the thousands separator: 1’234.56. This is not a cosmetic question. Reading a German datasheet with British eyes can cost you a factor of a thousand in the wrong direction, and being able to set the radix and the integer and fractional separators independently means the display can match the document on the bench rather than the other way around.

Grouping

The Indian numbering system groups differently again. One crore is 1,00,00,000, with groups of two after the first three. Big numbers in Indian financial reports look very different from big numbers in American ones, and the C47 and R47 don’t try to hide that; the Indian regional preset sets it up correctly.

Dates

Dates are worse. The US convention is MM/DD/YYYY, which the rest of the English-speaking world finds genuinely confusing. Most of the UK uses DD/MM/YYYY in everyday writing. Continental Europe writes DD.MM.YYYY. The ISO 8601 standard, YYYY-MM-DD, is the only one that sorts correctly when used as a filename or a column heading, which is why the calculator’s default is YMD. Once you’ve been bitten by 03/04/2025 (March or April?) you tend to settle into ISO whether you grew up with it or not.

Calendar

The Gregorian calendar was adopted at different times in different places. Catholic Europe switched in 1582. Britain and its colonies held out until 1752, with eleven days missing from that September. Japan adopted it in 1873. Russia waited until 1918. This sounds like an antique detail until you try to work out the day of the week for a historical event, at which point the right answer depends on which calendar the country was using at the time. The C47 and R47 know about this. Each regional preset sets the "first Gregorian day" appropriately, so that calendar arithmetic gives the locally correct answer rather than a tidy fiction.

Weeks

The "week of year" question gets a different answer depending on where you are. The US and India start the week on Sunday. Most of Europe and the UK start it on Monday. The ISO rule (week 1 contains the first Thursday) is what most European business reports use. American payroll systems generally use the Sunday-Saturday rule. The calculator handles both, and the week of year can be put in the status bar if you need to keep an eye on it.

Units

Even within units that look the same in two countries, the values aren't. A US gallon is 3.785 litres. An imperial (UK) gallon is 4.546 litres. A US fluid ounce is 29.6 mL, the imperial one is 28.4 mL. The CONVus flag controls which set the calculator reaches for first. This matters whenever fuel economy, recipes, or any liquid measure crosses the Atlantic.

Presets

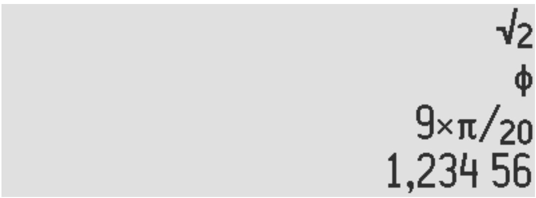
For most users the practical answer is the bottled regional preset on DISP page 3. CHINA, EUROP, INDIA, JAPAN, UK, USA, or DEFLT (which leans ISO). One keypress sets the radix, the separators, the date format, the calendar transition, and the week-of-year rule together. From there you can override individual settings if you have strong views on any of them. Most people set the preset once and forget about it. But it's worth knowing what it does, because it does quite a lot.

A tour around the disciplines

Tour

Rather than walking through specific values, here is the sort of thing different users tend to do with all this. The sketches that follow are illustrative rather than prescriptive, and certainly not a description of anybody in particular; nobody fits neatly into one column, and most of us borrow happily from several at once. The point is the breadth, not the destination.

Maths



The pure mathematician wants the irrational fractions mode on. With it, π and √2 and ϕ appear in their own clothes. Pair that with proper fractions, a sensible tolerance, and whichever display mode keeps the screen honest, and the calculator stops pretending that 1.732… is more honest than √3, or vice versa.

Physics

The physicist asks for a wide exponent range, leaves SPCRES on so that limits and infinities pass through quietly, and chooses the Physics convention for spherical coordinates (polar angle from the z-axis). HIDE is a useful friend here too: it stops genuine numerical zeros from advertising themselves as 10^{-31}.

Astronomy

The astronomer keeps the default exponent range because they can actually use it. Scientific notation, usually. Digits at a level that respects the data. Date arithmetic matters more here than for most users, so the regional preset and the calendar transition date earn their keep. Astronomers also tend to want the radix dot and not the comma, no matter where they live, because that's how the papers are written.

Chemistry

The chemist lives in UNIT mode. SI prefixes turn 100E-6 into 100µ without thinking. PFX.All extends the range out to quecto and quetta if you really need it, and most people don't, until they do. Significant figures actually mean something in the lab, so SIG or SCI mode tends to come into its own here.

Mechanical

The mechanical engineer tunes the fractions for the shop floor rather than the blackboard. DMX 32 gives sixteenths and thirty-seconds for inch work, DMX 12 gives the carpenter's quarters and eighths, and DMX 120 quietly admits $^{355}/_{113}$ when π comes up. Mixed fractions are nearly always on. The CONV menu can read either way around, depending on which side of the conversion you think of as the answer.

Civil

The civil engineer wants something between the mechanical and the financial: clean rounded answers, usually FIX or SIG to a sensible number of digits, and a status bar that shows where they are. The rounding mode matters when you're sizing a beam and "round half even" is not the same thing as "round up to be safe". The CONV menu pulls its weight here too. It carries a substantial library of named unit conversions (length, area, volume, force, pressure, energy and so on), and a single button press takes a value from kilonewtons to kips or metres to feet without the user having to remember the factor.

Surveying

The surveyor reaches for the angular and coordinate flags. Polar or rectangular. xyz or ijk display for vectors. The angular mode symbol can be made visible in the status bar, which has saved more than one of us from a degrees-versus-radians embarrassment. The clockwise-versus-anti-clockwise flag is worth a mention but a narrow one: it applies to plotting, where a coordinate pair can be drawn as a vector either anti-clockwise from the positive x-axis (mathematical) or clockwise from the positive y-axis (navigation). It doesn’t change how general calculations are done.

Electronics

The electronics engineer usually picks j over i, sets the multiplication symbol explicitly, and prefers 1+j×2 to 1+2j. UNIT mode handles the SI prefix work, and binary prefixes are one flag away when memory sizes turn up.

Power

The power engineer is a close cousin of the electronics engineer, but lives a few orders of magnitude up. Megawatts and Mvar rather than milliamps. ENGOVR for large reals, so the display jumps cleanly from kilo to mega to giga rather than wandering through 1.5×10^{6}. European nameplate conventions for the radix and separators if the work is on continental kit, US conventions if it isn’t.

Finance

The financial analyst goes the other way on most of this. Fractions off. FIX 2. End-of-period payments. A regional preset picks the radix and grouping that the currency expects. The calculator stops showing off and starts looking like a tape adder, which is the point. For one specific case, there’s an HP12C compatibility flag that rounds the periodic payment to two decimals before computing the amortisation schedule, so the result matches a printed bank statement to the last cent. It only affects amortisation, not the rest of the TVM solver, and the seasoned HP12C user generally wants it on so the calculator behaves the way they expect.

Statistics

The statistician sets up the plotting flags for the marker style and line behaviour they like, decides whether axes auto-scale together, and picks a digit count that doesn’t pretend to more precision than the data deserves. The C47 and R47 carry a fair selection of linear regression models (linear, logarithmic, exponential, power, and a few more) and a substantial library of probability distributions with their PDFs, CDFs and inverses; it’s a more complete statistical toolkit than the menu name suggests.

Coding

The computer scientist lives near the BASE menu, with the A-F keys repurposed as hex digits, leading zeros switched on, and the carry and overflow indicators in the status bar. Binary prefixes get used because memory comes in powers of 1024 and pretending otherwise is exhausting.

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